# AYA TLIJANI

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## PROFILE

Engineering student in Computer Science, specializing in **Data Science and Artificial Intelligence**, with experience in **machine learning research**, **RAG systems and LLM agents**, **data analysis** and **software development**. Seeking opportunities in data science, artificial intelligence and software development.

## PROFESSIONAL EXPERIENCE

**University of East London – Research Laboratory** **Jul 2026 – Nov 2026**
*Junior Research Assistant* *Remote*

**GNN-Based Reinforcement Learning for Project Scheduling** | Python, PyTorch, GNNs, Deep Reinforcement Learning

- Developed **graph neural network** policies using **PPO** and self-labeling to solve the Resource-Constrained Project Scheduling Problem (**RCPSP**), benchmarking against classical scheduling heuristics.
- Investigated LFT-anchored learning and conducted reproducible experiments across **PSPLIB** instances with up to 120 activities, statistical testing, and comparative evaluation.
- **Co-authored a research paper** examining fair benchmarking and learned scheduling policies, currently under review.

**AMSYS Consulting** **Aug 2026 – Dec 2026**
*AI Engineering Intern*

AI Research & Development, Computer Vision, Deep Learning, Generative AI

- Actively contributing to **AI research and development**, working on advanced **computer vision** and **deep learning** solutions.
- Supporting the development, experimentation, and evaluation of AI models and intelligent systems.

**ESPRIM – I5-Assist** **Jul 2026 – Aug 2026**
*Data Science Intern* *Monastir*

**Learning Analytics & Competency Assessment** | Python, NLP, Machine Learning, scikit-learn, Streamlit

- Developed an **NLP pipeline** using **text classification** and **weak supervision** to assess student competencies from AI interaction logs.
- Built an **interactive dashboard** for competency assessment, evidence-based scoring, student profiling, and learning progression analysis.

**GCT Gafsa** **Jul 2025 – Aug 2025**
*Introductory Internship* *Gafsa*

IT support, exploration of the company's network infrastructure, and participation in the development of an e-commerce platform with PrestaShop.

## PROJECTS

**SciLit** – *RAG System for Verified Literature Reviews* | Python, PyTorch, Hugging Face, RAG, LLMs, FastAPI GitHub

- Built an end-to-end **RAG system** over 1,109 NLP papers, combining **hybrid retrieval** (BM25 + dense embeddings, cross-encoder reranking) with a local **LLM** that writes cited literature reviews.
- Added automatic **fact-checking** of every generated sentence against the source papers (NLI): **95%** of sentences pass; fine-tuned **SciBERT** for sentence classification (0.749 macro-F1).

**QueryPilot** – *Self-Correcting Text-to-SQL Agent* | Python, LangGraph, LLMs, SQL, MCP GitHub

- Built a **LangGraph**-based **LLM agent** that generates SQL from natural-language questions, executes queries, and automatically diagnoses and repairs failures.
- Evaluated on 200 **BIRD Mini-Dev** questions across 11 databases using Qwen2.5-Coder 7B and Gemini API; improved execution accuracy from **47% to 51%** with Qwen and from **62.5% to 64%** with Gemini.
- Implemented a **read-only SQL sandbox**, tested against 16 write and injection attacks, and exposed database tools to other AI assistants through an **MCP** server.

**SmartRoute** – *Delivery Management & Route Optimization* | React, Python, FastAPI, OR-Tools, SQLite GitHub

- Developed a full-stack delivery dispatch platform featuring **VRPTW route optimization**, automated driver assignment, Excel data validation, real-time delivery tracking, and cash reconciliation.
- Built **REST APIs**, role-based authentication, interactive dashboards, and delivery analytics with automated end-to-end testing.

## EDUCATION

**ESPRIT Monastir (ESPRIM)** **Sep 2024 – Present**
*Engineering Degree in Computer Science, specializing in Data Science and Artificial Intelligence* *Monastir*

**Preparatory Institute for Engineering Studies of Monastir (IPEIM)** **Sep 2021 – Jun 2024**
*Preparatory Institute for Engineering Studies* *Monastir*

## SKILLS, CERTIFICATIONS & LANGUAGES

- **Languages:** Python, SQL, Java, JavaScript, PHP, C, C#
- **Machine Learning & AI:** Deep Learning, Deep Reinforcement Learning, Graph Neural Networks, NLP, Computer Vision, Generative AI, PyTorch, PyTorch Geometric, TensorFlow, scikit-learn
- **LLMs, RAG & Agents:** LangGraph, MCP, Ollama, Hugging Face Transformers, FAISS, BM25, Text-to-SQL, LLM Evaluation
- **Optimization:** OR-Tools, Vehicle Routing (VRPTW), Project Scheduling (RCPSP)
- **Web & Backend:** FastAPI, REST APIs, React, Angular, Symfony, JavaFX, HTML/CSS
- **Data & Tools:** Pandas, NumPy, SQLite, Streamlit, Power BI, Databricks, Jupyter, Git, Docker, pytest
- **Certification:** NVIDIA Deep Learning Institute – Fundamentals of Deep Learning
- **Languages:** Arabic (Native), English (C1), French (B2), Spanish (Basic)